



Exercise Session 2

📅 29.09.2025

Algorithms & Data Structures HS 25 G-21

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Plan for Today

1. Moodle Quiz

- Moodle quiz
- Moodle quiz discussion

2. Organization

- Exercise website

3. Homework

- Discussing exercises from Exercise Sheet 1
- Main focus Ex. 1.2 and Ex. 1.3

4. Theory recap

- O – notation
- Star search (if we have time)

5. Code Expert demo

6. In Class Exercises

- O -notation exercises
- Mathematical Induction

For $n \in \mathbb{N}$, let $f(n) = n^2 + 1001n + n^3$ and $g(n) = 10n^3$. Then $f(n)$ has the same asymptotic growth rate as $g(n)$ (meaning that $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} \in \mathbb{R}_+$).

☐ True

☐ False

For $n \in \mathbb{N}$, let $f(n) = n^2 + 1001n + n^3$ and $g(n) = 10n^3$. Then $f(n)$ has the same asymptotic growth rate as $g(n)$ (meaning that $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} \in \mathbb{R}_+$).

☒ True

☐ False

Solution: We calculate $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)}$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{n^2 + 1001n + n^3}{10n^3} = \lim_{n \rightarrow \infty} \frac{1}{10n} + \frac{1001}{10n^2} + \frac{1}{10}$$

$$= \frac{1}{10} \in \mathbb{R}^+ \Rightarrow \text{the statement is true.}$$

$$n^4 \leq O\left(\frac{n^4}{\log(n)}\right)$$

☒ True

☐ False

$$n^4 \leq O\left(\frac{n^4}{\log(n)}\right)$$

☐ True

☒ False

Solution: We know from Theorem 1, that
 $n^4 \leq O\left(\frac{n^4}{\log(n)}\right)$ if $\lim_{n \rightarrow \infty} \frac{\frac{n^4}{\log(n)}}{\frac{n^4}{\log(n)}} = 0$ or C , $C \in \mathbb{R}^+$

We calculate $\lim_{n \rightarrow \infty} \frac{\frac{n^4}{\log(n)}}{\frac{n^4}{\log(n)}}$

$$\lim_{n \rightarrow \infty} \frac{\frac{n^4}{\log(n)}}{\frac{n^4}{\log(n)}} = \lim_{n \rightarrow \infty} \log(n) = \infty \Rightarrow \text{from Th. 1 we know } n^4 \not\leq O\left(\frac{n^4}{\log(n)}\right)$$

$$e^{3\ln(n)} \leq O(n^2)$$

☐ True

☐ False

$$e^{3\ln(n)} \leq O(n^2)$$

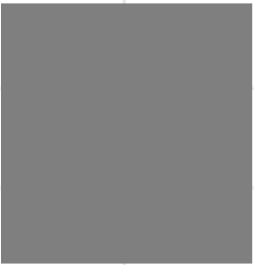
☐ True

☒ False

We want to use Th. 1 so we calculate $\lim_{n \rightarrow \infty} \frac{e^{3\ln(n)}}{n^2}$

$$\lim_{n \rightarrow \infty} \frac{e^{3\ln(n)}}{n^2} = \lim_{n \rightarrow \infty} \frac{n^3}{n^2} = \lim_{n \rightarrow \infty} n = \infty \Rightarrow \text{from Th. 1 we know } e^{3\ln(n)} \neq O(n^2)$$

Let $f(n) = 6n^2 + 5n + 10$ (for $n \in \mathbb{N}$). For each of the following definitions of $g(n)$, is $g(n) \leq O(f(n))$?

True	False	
		$g(n) = 123456789n - 200\sqrt{n}$
		$g(n) = 0.01n^2 \log(n)$
		$g(n) = 10n^3 + 5n + 1000$

Scoring method: **Subpoints** ?

Let $f(n) = 6n^2 + 5n + 10$ (for $n \in \mathbb{N}$). For each of the following definitions of $g(n)$, is $g(n) \leq O(f(n))$?

True	False		
☒	☐	$g(n) = 123456789n - 200\sqrt{n}$	
☐	☒	$g(n) = 0.01n^2 \log(n)$	
☐	☒	$g(n) = 10n^3 + 5n + 1000$	

Scoring method: **Subpoints** ?

We will use Th 1. $\Rightarrow$ we calculate $\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)}$

$$g(n) = 123456789n - 200\sqrt{n}$$

$$\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{123456789n - 200\sqrt{n}}{6n^2 + 5n + 10}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 (123456789 \cdot \frac{1}{n} - 200 \cdot \frac{1}{n\sqrt{n}})}{n^2 (6 + \frac{5}{n} + \frac{10}{n^2})} = 0 \Rightarrow \text{from Th. 1 we know } g(n) \leq O(f(n))$$

$$g(n) = 0,01n^2 \log(n)$$

$$\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{0,01 \cdot n^2 \log(n)}{6n^2 + 5n + 10} = \lim_{n \rightarrow \infty} \frac{n^2 (0,01 \cdot \log(n))}{n^2 (6 + \frac{5}{n} + \frac{10}{n^2})}$$

$$= \lim_{n \rightarrow \infty} \frac{0,01 \cdot \log(n)}{6} = \infty \Rightarrow \text{from Th. 1 we know } g(n) \neq O(f(n))$$

$$g(n) = 10n^3 + 5n + 100$$

$$\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{10n^3 + 5n + 100}{6n^2 + 5n + 10} = \lim_{n \rightarrow \infty} \frac{n^2 (10n + \frac{5}{n} + 100 \cdot \frac{1}{n^2})}{n^2 (6 + \frac{5}{n} + \frac{10}{n^2})}$$

$$= \lim_{n \rightarrow \infty} \frac{10n}{6} = \infty \Rightarrow \text{from Th. 1 we know } g(n) \neq O(f(n))$$

Let $f : \mathbb{N} \rightarrow \mathbb{R}_+$ be some function for which we would like to prove $f(n) \geq n^2$ for every $n \geq 1$. Assume that you have proven that:

- $f(2) \geq 2^2$
- If $f(k) \geq k^2$ holds for an arbitrary positive integer k , then $f(k+1) \geq (k+1)^2$ holds.

Then, $f(n) \geq n^2$ holds for all positive integers $n \geq 1$.

☒ True

☐ False

Let $f : \mathbb{N} \rightarrow \mathbb{R}_+$ be some function for which we would like to prove $f(n) \geq n^2$ for every $n \geq 1$. Assume that you have proven that:

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Then, $f(n) \geq n^2$ holds for all positive integers $n \geq 1$.

☐ True

☒ False

Solution: The statement that $f(n) \geq n^2$ holds for all positive integers $n \geq 1$ is false, since the Base case is for $n=2$ and with the induction step the claim is proved for values of n bigger than 2. $\Rightarrow$ It is never shown that the inequality holds for $n=1$.

Theory recap > O - notation

Asymptotic notation (O-notation)

Notation: $N = \{n_0, n_0+1, \dots\}$ - the set of all possible arguments of the function f . (In our case we consider f to be the function that gives us the runtime of

an algorithm and therefore N is a set of all possible input sizes.)! N is not necessarily $\mathbb{N}$ (the set of the natural numbers). It is possible that $N = \mathbb{N}$, but this is not always true. N is just a set of all possible inputs of f .

Definition: For $f: \mathbb{N} \rightarrow \mathbb{R}^+$, we define

$$O(f) := \{g: \mathbb{N} \rightarrow \mathbb{R}^+ \mid \exists C \in \mathbb{R}^+ : \forall n \in \mathbb{N} : g(n) \leq C \cdot f(n)\}$$

Definition (with words) For a function f with domain N (domain are the possible values of the argument of f) and codomain $\mathbb{R}^+$ (codomain is the set of values of $f(x), x \in N$) ($\mathbb{R}^+$ are the positive real numbers) we define $O(f)$. $O(f)$ is defined as the set of all functions g with domain N and codomain $\mathbb{R}^+$ for which there exists a positive real constant C , such that for every n in the domain N it holds: $g(n) \leq C \cdot f(n)$ (the value g takes in point n is less or equal to the value of f in the same point n multiplied by the constant C).

!Important: the constant C is the same for all $n \in N$

Theorem 1. Let N be an infinite subset of $\mathbb{N}$ and $f : N \rightarrow \mathbb{R}^+$ and $g : N \rightarrow \mathbb{R}^+$.

- If $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$, then $f \leq O(g)$ and $g \not\leq O(f)$.
- If $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = C \in \mathbb{R}^+$, then $f \leq O(g)$ and $g \leq O(f)$.
- If $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$, then $f \not\leq O(g)$ and $g \leq O(f)$.

The following theorem can also be helpful when working with O -notation.

Theorem 2. Let $f, g, h : N \rightarrow \mathbb{R}^+$. If $f \leq O(h)$ and $g \leq O(h)$, then

1. For every constant $c > 0$, $c \cdot f \leq O(h)$.
2. $f + g \leq O(h)$.

Notice that for all real numbers $a, b > 1$, $\log_a n = \log_a b \cdot \log_b n$ (where $\log_a b$ is a positive constant). Hence $\log_a n \leq O(\log_b n)$. So you don't have to write bases of logarithms in asymptotic notation, that is, you can just write $O(\log n)$. Be careful though, as this does not apply when the logarithms are in some exponent! For example, $e^{\ln(n)} = n$ but $e^{\log(n)} = n^{\log e}$, and these two expressions are certainly not of the same order.

CodeExpert Demo

In the Exercises page you can find all exercises – complete, incomplete, ones you need to do and those for which submission is closed and solutions are published.

Autumn 2024

Algorithmen und Datenstrukturen

Change course ▾

Exercises

Code examples

My exercise group

Analytics

In the Analytics page you can find statistics of how many people managed to collect points within a particular range and in which group you belong.

Exercises

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To do



Incomplete



Complete



Feedback

Warm-Up Assignment



Welcome



Median of Three

Programming Assignment 1

6 XP



Subarray with Few Negative Elements

3 XP



Minimum Changes to Uniform

5 XP

Programming Assignment 2

6 XP

To do – new tasks you need to complete. There are only tasks you have not made a submission to. If you have submitted a solution to a task even if it did not get the full amount of points it is moved from To do.

When is the deadline for submission of your solutions for the particular set of problems.

Closed 1 year ago

Solutions available

In Incomplete you can find the exercises to which you have submitted a solution, but your solution did not get the full amount of points.

Closed 11 months ago

Solutions available

In Complete you can find the exercises to which you have submitted a solution and your solution got the full amount of points.

Closed 11 months ago

Solutions available

Neli Tsokanova



Всички отметки



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To do Incomplete Complete Feedback

Warm-Up Assignment

Welcome

Median of Three

Programming Assignment 1

Subarray with Few Negativ

Minimum Changes to Unifo

Programming Assignment 2

6 XP

When you select a task you want to solve a window appears on the right and when you click on Edit in IDE, the CodeExpert IDE opens. There you can find the problem description as well as the code template you need to fill in.

Warm-Up Assignment

Welcome

Closed 1 year ago

Edit in IDE View solution

Ungraded task

This task is not graded, therefore all submissions are accepted.

Submissions

> Sun 6.10.2024, 20:03 100%	View submission
> Sun 6.10.2024, 20:00 0%	View submission

Reset task ...

Here you can also find your submissions for the task, your score for the particular submission. Refer to the next slides to see what happens when you click View submission.

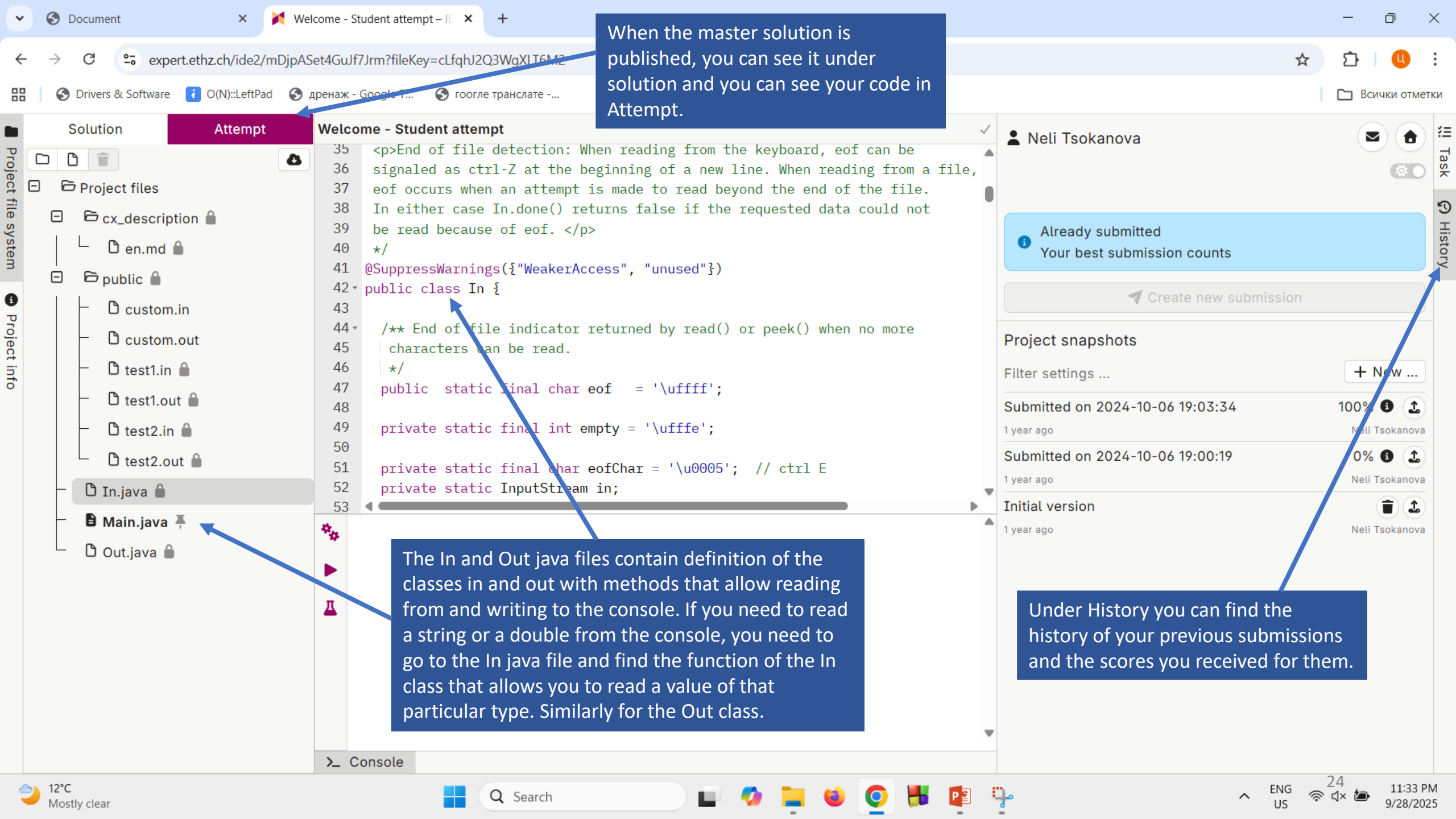
12°C Mostly clear

Search

ENG US

21

10:54 PM 9/28/2025



When the master solution is published, you can see it under solution and you can see your code in Attempt.

The In and Out java files contain definition of the classes in and out with methods that allow reading from and writing to the console. If you need to read a string or a double from the console, you need to go to the In java file and find the function of the In class that allows you to read a value of that particular type. Similarly for the Out class.

Under History you can find the history of your previous submissions and the scores you received for them.

Document

Exercises | ETH Code Expert

Welcome - Submission feedback

expert.ethz.ch/ide2/WNsHRZXLwJRTzemmE

Drivers & Software

O(N)::LeftPad

дренаж - Google T...

гоогле транслате -...

Всички отметки

Solution

Feedback

Welcome - Submission feedback

Read-Only

Project file system

Project info

Project files

cx_description

en.md

public

custom.in

custom.out

test1.in

test1.out

test2.in

test2.out

In.java

Main.java

Out.java

1

class Main {

2

public static void main(String[] args) {

3

// Uncomment the following two lines if you want to read from a file

4

// In.open("public/test1.in");

5

// Out.compareTo("public/test1.out");

6

7

int testNum = In.readInt(); // read the number of tests

8

for (int i = 0; i < testNum; i++) {

9

String s = In.readWord();

10

Out.println(welcome(s));

11

}

12

13

// Uncomment this line if you want to read from a file

14

// In.close();

15

}

16

17

public static String welcome(String s) {

18

return "Welcome " + s + "!";

19

}

20

Compilation successful

The time limit for this exercise is 2 seconds.

Testing solution..

Test set 1: Correct answer (0.001)

Test set 2: Correct answer (0.0)

Total score: 100 / 100

Console

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Feedback

This task is not graded

Your submission was not manually reviewed.

This is the window you see when you click View Submission. Here you can see the score you got on the hidden tests.

12°C

Mostly clear

Search

ENG US

25

11:42 PM

9/28/2025

In Class Exercises

Level 1.

Claim	True	False
1) $f(n) = 3n + 5$, $g(n) = n^2$, $f \leq O(g)$		
2) $f(n) = 7n^2 + 4n + 1$, $g(n) = n^2$, $f \leq O(g)$		
3) $f(n) = n^2 + \log n$, $g(n) = n^3$, $f \neq O(g)$		
4) $f(n) = 50 \log n$, $g(n) = n$, $f \leq O(g)$		
5) $f(n) = n \log n$, $g(n) = n^{1.5}$, $f \neq O(g)$		

Level 2

15/17

	Claim	True	False
1)	$f(n) = n^3, g(n) = 2^n, f \leq O(g)$		
2)	$f(n) = 2^n, g(n) = 3^n, f \leq O(g)$		
3)	$f(n) = \sqrt{n}, g(n) = \ln^2 n, f \leq O(g)$		
4)	$f(n) = n \log(n), g(n) = n \cdot (\log n)^2, f \leq O(g)$		
5)	$f(n) = n^{\ln(n)}, g(n) = 2^n, f \leq O(g)$		

Level 3 (Try yourselves)

	Claim	True	False
1)	$f(n) = n^{1.1}$, $g(n) = n \log n$, $f \leq O(g)$		
2)	$f(n) = 2^{\frac{n}{\ln n}}$, $g(n) = 2^{n^{0.9}}$, $f \leq O(g)$		
3)	$f(n) = n^{e n}$, $g(n) = e^{\sqrt{n}}$, $f \leq O(g)$		

Mathematical Induction Exercises

Exercise 1 The sequence $\{a_n\}$ is given with the recursive formula:

$$a_1 = 1$$

$$a_n = \frac{a_{n-1}}{a_{n-1} + 1}$$

Find a formula for a_n in terms of n .

Exercise 2 Prove the equality for $\forall n \in \mathbb{N}$

$$2 + 4 + 15 + \dots + (n^2 + 2n - 1) = \frac{n(2n^2 + 9n + 1)}{6}$$

Exercise 3 Prove the inequality for $\forall n \in \mathbb{N}$

$$\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{n+n} > \frac{13}{24}$$

Thank you for your attention
and participation!